

Open source solutions used by ITM

- Servers: CentOS, Fedora Core 5, Fedora 8, LTSP, Open LDAP, Free RADIUS, Free NAS and CUPS.
- Desktops: Fedora and Ubuntu
- Firewall and proxy: Shorewall and Squid
- Network monitoring: NTop and MRTG
- Systems administration: Webmin
- Databases: MySQL and PostgreSQL
- Special applications: Joomla, Asterisk and Moodle
- Desktop applications: OpenOffice.org, Thunderbird, Firefox, GIMP, VLC, etc.

the staff and faculty—a few of which are thin clients-cum-print servers. The number of laptops available for students is over 400.

Connecting people: networking security

The campus is divided into two independent networks. The wired network is used mainly by staff and faculty, while the wireless network is dedicated for students. Only the data exchange server, the Prowess database and intranet server have data common to the faculty and students. In each of these servers, the institute has two network cards that provide connectivity to each group while keeping the networks independent. All the servers—including the DNS, DHCP, LDAP, the Web server (intranet), LTSPs, NFS, the data exchange server and the NAS box (back-up server)—are in the staff network, which is completely isolated from the students.

To provide the much needed and efficient connectivity among the servers and clients, structured cabling is in place, connecting two server rooms located at diagonally opposite corners of the campus. The server rooms house the switch racks, with cables running directly to each machine on the campus or one of the many Wi-Fi access points. There are 18 access points by NetGear with RADIUS authentication deployed to meet all requirements.

Security

This kind of widespread network needs a lot of security measures—some physical and some at the virtual level. Besides physical security, the institute also ensures the normal log-in access control to servers.

The most important measure to secure servers is at the physical level, with access allowed only to authorised people. All servers are housed in the two server rooms, which normally remain locked and the keys are only with the senior systems administrator during each of the two shifts.

For login-based access control, only senior systems administrators have passwords for root access and junior administrators have limited access (using *sudo*) to servers. All systems administrators have to login with their own user names and associated passwords.

Normal users have no administrative access to any of the servers.

All users, including administrators, are authenticated through LDAP servers for network access. Administrators are assigned different privileges on the server by the senior systems administrator. All cannot work on all servers. Faculty and staff have access only to data in their home directories, Prowess database and to common shared data in the data exchange server. The faculty and staff can view data on the intranet server.

All the users in both the networks (staff, faculty and students) are authorised users with different levels of authorisation. The LTSP users (staff and faculty) are authenticated centrally through the LDAP server. All students are centrally authenticated through the RADIUS server. Also, for wireless access, MAC address bindings are used to assign the IP address to the authorised students only.

Security from intruders

The gateway and proxy servers that connect to the Internet run the Shoreline firewall (commonly known as Shorewall). All ports on these two machines are closed on the public interface. Even local internal servers normally have only need-based ports open. Besides all the above restrictions, the ERP, Web and mail servers allow SSH access only from one IP address.

Training day

Setting up this kind of an infrastructure that uses a lot of free software solutions needed a lot of expertise. The institute initially hired a consulting company. The primary role of the company was to set up the LTSP server and deploy around 20 thin clients.

This deployment also meant a learning process. The staff members needed to be trained in using this new technology. The secondary responsibility of the company was to train the systems administrators who later became trainers themselves. It was now mandatory for all systems administrators to either be certified before joining, or obtain Red Hat Certification within one year.

All training for staff and faculty was done in-house by systems administrators. The institute is now planning to use its experience to train outsiders so that others can also reap the benefits of Linux and Free Software technologies. 

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A Free Software fund-a-mental-ist and Charles Bukowski fan, Swapnil also writes fiction and tries to find cracks in a proprietary company's 'paper armours'. He is a big movie buff, and prefers listening to music at such loud volumes that he's gone partially deaf when it comes to identifying anything positive about proprietary companies. Oh, and he is also the assistant editor of *EFYTimes.com*.